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```
clc;
clear;
close all;
```

PARAMETERS

```
M = 16;
k = log2(M);
```

```
Nfft = 64;
Nofdm = 100;
```

TRANSMITTER

```
% Generate random bits
Nbits = Nofdm * Nfft * k;

txBits = randi([0 1], Nbits, 1);

% 16-QAM modulation
txSymbols = qammod(txBits, M, ...
    'InputType', 'bit', ...
    'UnitAveragePower', true);

% Arrange QAM symbols into OFDM symbols
txSymbolsMatrix = reshape(txSymbols, Nfft, Nofdm);

% IFFT
txOFDM = ifft(txSymbolsMatrix, Nfft);
```

CHANNEL

```
% No noise or fading
rxOFDM = txOFDM;
```

RECEIVER

```
% FFT
rxSymbolsMatrix = fft(rxOFDM, Nfft);

% Convert to serial
rxSymbols = rxSymbolsMatrix(:);

% 16-QAM demodulation
rxBits = qamdemod(rxSymbols, M, ...
    'OutputType', 'bit', ...
    'UnitAveragePower', true);
```

BER

```
[numErrors, ber] = biterr(txBits, rxBits);

fprintf('Number of bit errors = %d\n', numErrors);
fprintf('BER = %f\n', ber);

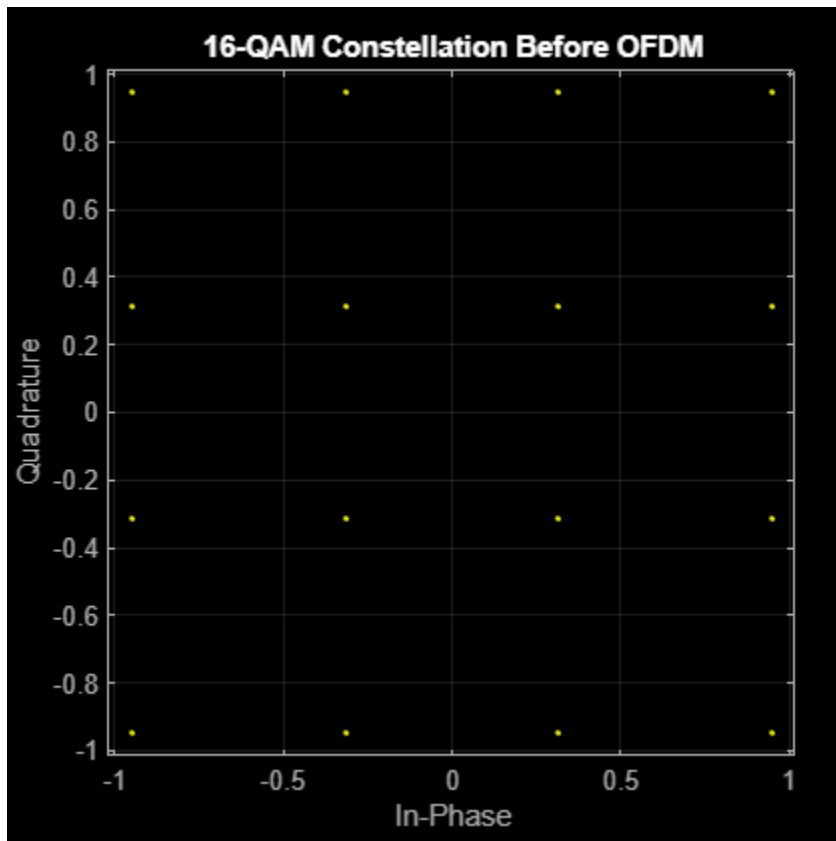
Number of bit errors = 0
BER = 0.000000
```

PLOT 1: TRANSMITTED CONSTELLATION

```
figure;

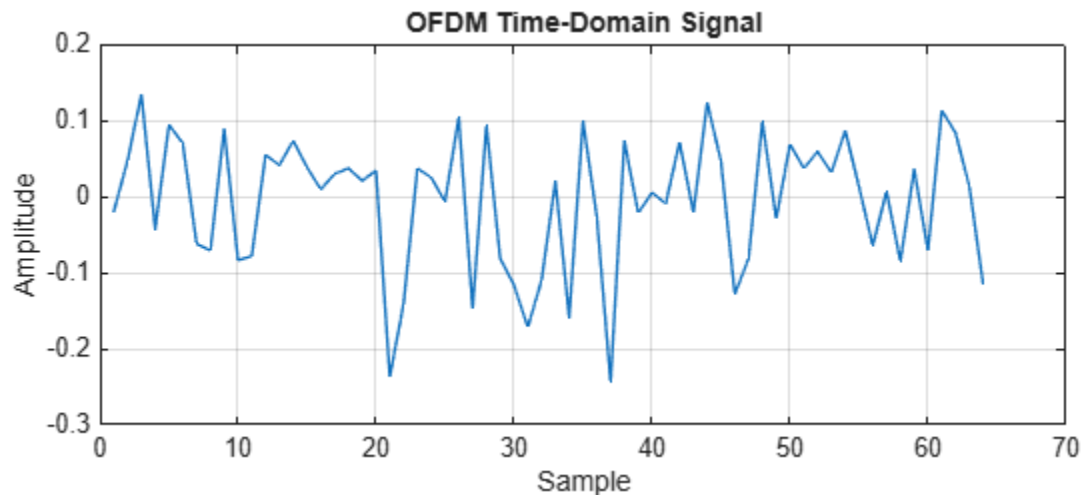
scatterplot(txSymbols);

title('16-QAM Constellation Before OFDM');
grid on;
```



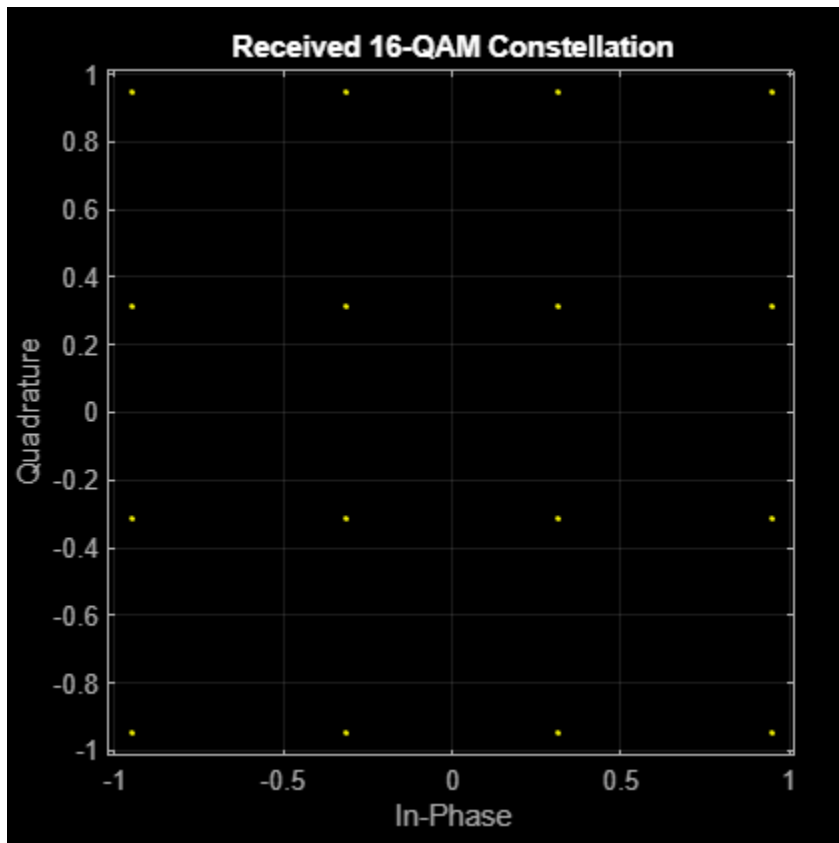
PLOT 2: OFDM TIME-DOMAIN SIGNAL

```
figure;  
plot(real(txOFDM(:,1)));  
xlabel('Sample');  
ylabel('Amplitude');  
title('OFDM Time-Domain Signal');  
grid on;
```



PLOT 3: RECEIVED CONSTELLATION

```
figure;  
scatterplot(rxSymbols);  
title('Received 16-QAM Constellation');  
grid on;
```



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